4 Multiple Integrals

4.4 Applications of Double Integrals

Definition 4.4.1 If D is a (completely flat) plate, and $f(x, y)$ denotes the mass D at the point (x, y), the total mass of D is simply

$$\iint_D f(x, y)dA.$$

Definition 4.4.2 Suppose that we have a (completely flat) lamina which occupies the region D and whose mass at point (x, y) is $\rho(x, y)$. Its center of mass is given by the point $(\overline{x}, \overline{y})$ where

$$m\overline{x} = \iint_D x\rho(x, y)dA, \quad \text{and} \quad m\overline{y} = \iint_D y\rho(x, y)dA,$$

where m is the total mass of the lamina.

Example 4.4.3 Find the total mass and center of mass of the lamina which occupies the triangular region of space with vertices $(0, 0)$, $(1, 0)$, and $(0, 2)$ and whose mass at point (x, y) is given by $\rho(x, y) = 1 + 3x + y$.

4.5 Surface Area

Example 4.5.1 Find the surface area of the surface

$$z = x^2 + 2y + 2$$

over the triangular region with vertices $(0, 0)$ $(1, 0)$ and $(0, 1)$.

Example 4.5.2 Find the surface area of the surface $z = x^2 + y^2$ which lies beneath the plane $z = 9$.